








Statistics of Financial Markets

Seminar 3: α-stable distributions



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CI**: Confidence Interval
- **CLT**: Central Limit Theorem
- **CMS**: Chambers–Mallows–Stuck (simulation method)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GCLT**: Generalised Central Limit Theorem
- **ML**: Maximum Likelihood
- **QQ**: Quantile–Quantile (plot)
- **SE**: Standard Error

Today's Question and Seminar Structure

- Question of the seminar
  - **can a stable law with $\alpha < 2$ describe daily returns, and what does it imply about their sums?**
- Exercises: in the order of the Exercise Map (next slide)
- Outputs to produce
  - the answers to the A exercises; the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Notebook for today: open the seminar notebook in Google Colab; each task names its section
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	how does the scale of a sum grow with the number of terms?	Solved, Proposed	A1
A3, A4	what changes between the S1 and S0 parameterisations?	Solved, Proposed	A3
A5, A6	how fast do power-law tails fall?	Solved, Proposed	A5
A7, A8	McCulloch estimates from five quantiles	Solved, Proposed	A7
A9, A10	one stable draw by the CMS method	Solved, Proposed	A9
B1, B2	how heavy are the tails of BET, S&P 500, DAX and Bitcoin returns?	Solved, Proposed	B1
B3, B4	ML fit, QQ plot, large losses: stable, Normal and Student-t	Solved, Proposed	B3
B5, B6	does α stay the same for weekly and monthly returns? does the variance settle?	Solved, Proposed	B5
C1, C2	is α constant over time? what is wrong in an AI answer?	Proposed	B1, B3

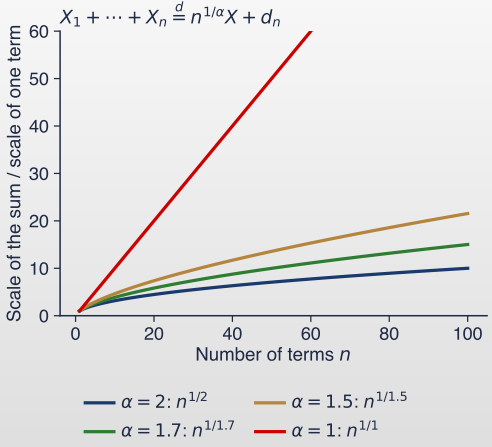
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model
- **A**: calculations on paper; **B**: real data, in the notebook; **C**: open questions and the critique of an AI answer

Data Used

Series	Source	Price used	Period
BET, S&P 500, DAX	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin)
- Weekly and monthly log returns: sums of the daily log returns of each calendar week or month
- In the notebook: `log_returns('bet', '2000-01-01')`; no account or key is needed

Prerequisites for Today (1/14): Stable Laws



- X is **stable** if, for independent copies X_1, X_2 and any $a, b > 0$

$$aX_1 + bX_2 \stackrel{d}{=} cX + d$$

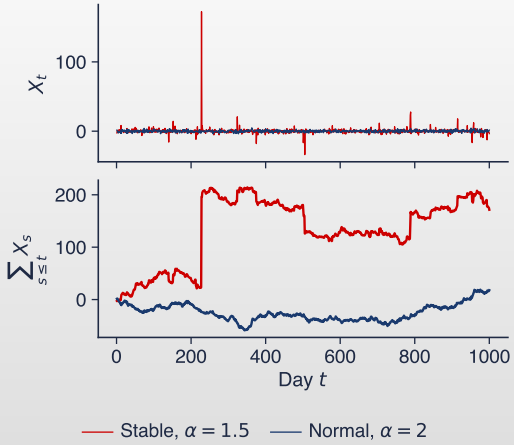
- $\stackrel{d}{=}$: has the same distribution as; a, b: weights; $c > 0$: the scale and d: the shift of the result
  - a weighted sum keeps the shape of one copy; only the scale and the location change
- **Scale rule**, with the **index of stability** $0 < \alpha \leq 2$

$$c^\alpha = a^\alpha + b^\alpha, \qquad X_1 + \cdots + X_n \stackrel{d}{=} n^{1/\alpha}X + d_n$$

- n i.i.d. copies: the scale of the sum is $n^{1/\alpha}$ times the scale of one term; d_n: a shift
  - Normal, $\alpha = 2$: $\sqrt{n}$; Cauchy, $\alpha = 1$: n, so the scale grows as fast as the sum
  - the smaller α, the faster the scale of a sum grows (chart)

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Prerequisites for Today (2/14): Stable Paths



- Two classical stable laws
  - Normal ($\alpha = 2$): variances add, $c^2 = a^2 + b^2$
  - Cauchy ($\alpha = 1$, density $1/(\pi(1 + x^2))$): scales add, $c = a + b$
- In the chart: 1000 simulated daily returns X_t (top) and their running sums (bottom)
  - blue: Normal, a **random walk**: the sum moves by small, similar steps
  - red: stable with $\alpha = 1.5$: rare huge jumps, and one jump can dominate the whole sum
- Mandelbrot (1963): cotton price changes look like the red path, not the blue one

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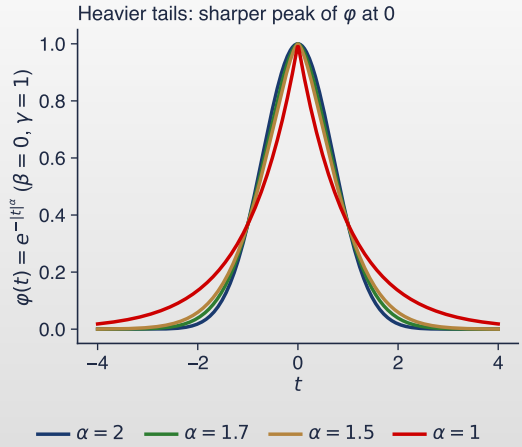
Prerequisites for Today (3/14): the Generalised Central Limit Theorem

- **GCLT** (generalised central limit theorem): for $X_1, \ldots, X_n$ i.i.d., if for some constants $a_n > 0$, b_n

$$\frac{X_1 + \cdots + X_n - b_n}{a_n} \xrightarrow{d} Z,$$

- then the limit Z is stable; $\xrightarrow{d}$: convergence in distribution
  - a_n rescales and b_n recentres the sum
- Which limit
  - finite variance: the Normal distribution, with $a_n \propto \sqrt{n}$ (the usual CLT)
  - power-law tails $P(|X| > x) \approx Cx^{-\alpha}$ with $\alpha < 2$: a stable law with index α, with $a_n \propto n^{1/\alpha}$
  - $\propto$: proportional to
- Consequence for returns
  - a weekly or monthly return is a sum of daily returns
  - if daily returns were i.i.d. stable, weekly and monthly returns would have the **same** α (tested in B5)

Prerequisites for Today (4/14): the Characteristic Function



- **Characteristic function** of X

$$\varphi(t) = E\left[e^{itX}\right] = E[\cos tX] + i\,E[\sin tX]$$

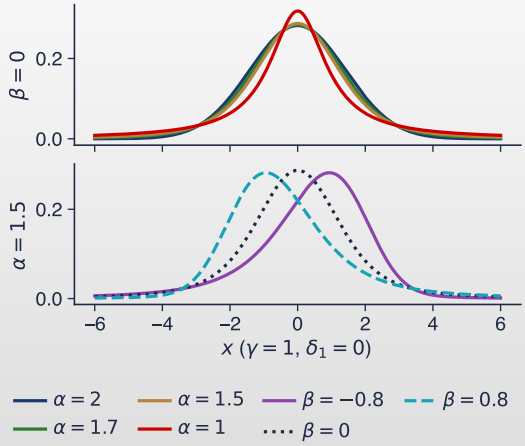
- $i = \sqrt{-1}$: the imaginary unit; t: a real argument; $E[\cdot]$: the expected value
  - it always exists ($|e^{itX}| = 1$) and determines the law completely
- Why it is used here
  - stable laws have no closed-form density in general, but a simple φ
  - for independent terms, $\varphi_{X_1+X_2}(t) = \varphi_{X_1}(t)\,\varphi_{X_2}(t)$: sums become products
- Symmetric case $\beta = 0$, $\gamma = 1$, $\delta = 0$ (chart)

$$\varphi(t) = e^{-|t|^{\alpha}}$$

- Smaller α: φ has a sharper peak at 0 and the density heavier tails

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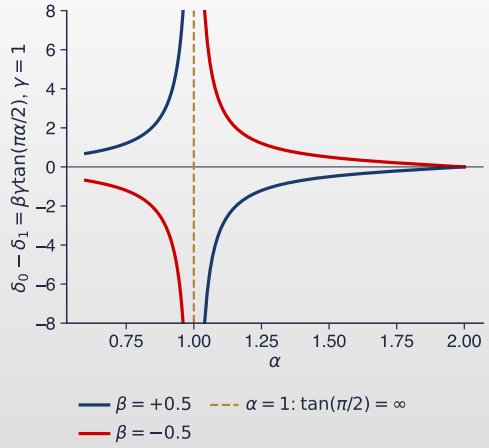
Prerequisites for Today (5/14): Four Parameters



- $S(\alpha, \beta, \gamma, \delta)$: the stable law with four parameters
- $\alpha \in (0, 2]$: **index of stability** (tail index); smaller α, heavier tails and a taller centre (top)
  - $\alpha = 2$: the Normal distribution; $\alpha = 1$: Cauchy
- $\beta \in [-1, 1]$: **skewness**
  - $\beta > 0$: a heavier right tail; $\beta < 0$: a heavier left tail; $\beta = 0$: symmetric (bottom)
- $\gamma > 0$: **scale**; δ: **location**
  - $\gamma X + \delta$ has scale γ and location shifted by δ
  - γ is not a standard deviation: for $\alpha < 2$ the variance is infinite
- The meaning of δ depends on the parameterisation (next slide)

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Prerequisites for Today (6/14): Parameterisations S1 and S0



- **S1**, for $\alpha \neq 1$

$$\ln \varphi(t) = -\gamma^{\alpha}|t|^{\alpha}\left[1 - i\beta\,\mathrm{sign}(t)\tan\frac{\pi\alpha}{2}\right] + i\delta_{\mathbf{1}}t$$

- **S0**, for $\alpha \neq 1$

$$\ln \varphi(t) = -\gamma^{\alpha}|t|^{\alpha}\left[1 + i\beta\,\mathrm{sign}(t)\tan\frac{\pi\alpha}{2}\left(|\gamma t|^{\mathbf{1}-\alpha} - 1\right)\right] + i\delta_{\mathbf{0}}t$$

- $\mathrm{sign}(t)$: $+1$ for $t > 0$, -1 for $t < 0$; the β term makes the law asymmetric
  - $|\varphi(t)| = e^{-\gamma^{\alpha}|t|^{\alpha}}$ in both: the shape is the same
- Only the location differs

$$\delta_{\mathbf{0}} = \delta_{\mathbf{1}} + \beta\gamma\tan\frac{\pi\alpha}{2}$$

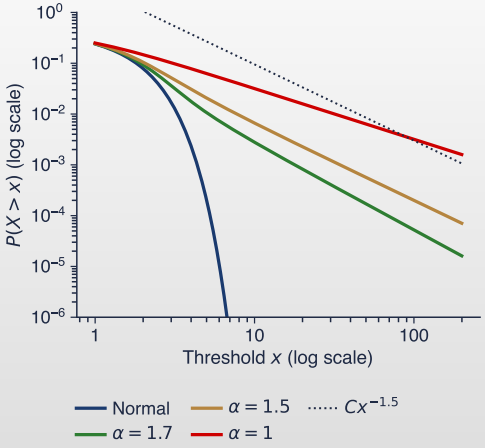
- $\delta_{\mathbf{0}}$, $\delta_{\mathbf{1}}$: the location in S0 and in S1
  - near $\alpha = 1$ the shift explodes (chart), so S0 is preferred for estimation

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Prerequisites for Today (7/14): Special Cases and Software

- $\alpha = 2$: the term with β vanishes, since $\tan \pi = 0$
  - $\varphi(t) = e^{-\gamma^2 t^2 + i\delta t}$, the characteristic function of $N(\delta, 2\gamma^2)$; the variance is $2\gamma^2$, not γ^2
  - compare: $N(\mu, \sigma^2)$ has $\varphi(t) = \exp(i\mu t - \sigma^2 t^2/2)$
- $\alpha = 1$, $\beta = 0$: the **Cauchy** law
  - $\varphi(t) = e^{-\gamma|t| + i\delta t}$; density $\gamma/(\pi(\gamma^2 + (x-\delta)^2))$; no mean, no variance
- **Software**: know the parameterisation before reading the numbers
  - `scipy.stats.levy_stable` uses S1 by default: `loc` $= \delta_1$, `scale` $= \gamma$
  - to compare with S0 results, convert δ_1 to δ_0 (Prerequisites for Today (6/14))

Prerequisites for Today (8/14): Power-Law Tails



- For $\alpha < 2$, the tails fall like a power of x

$$P(X > x) \approx C\, x^{-\alpha}, \qquad x \to \infty$$

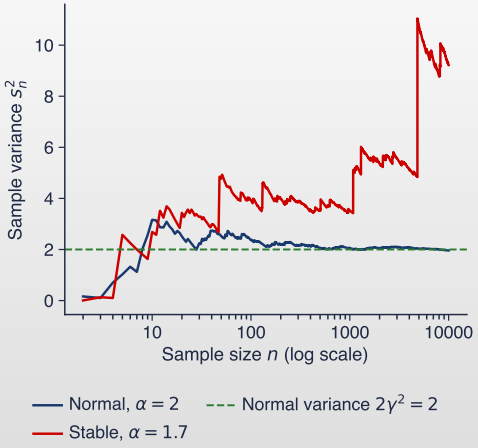
- $C > 0$: a constant; on a log-log scale the tail is a straight line of slope $-\alpha$
  - the Normal tail falls much faster, like $e^{-x^2/(2\sigma^2)}$ (blue curve)
- Ratio of two tail probabilities

$$\frac{P(X > kx)}{P(X > x)} \approx k^{-\alpha}$$

- doubling the threshold ($k = 2$) divides the probability only by 2^α
  - $\alpha = 1.5$: by $2^{1.5} = 2.83$; large losses stay frequent
  - "1 day in 50" above x becomes "1 day in $50 \cdot 2^\alpha$" above $2x$

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Prerequisites for Today (9/14): Which Moments Exist



- For $\alpha < 2$, the absolute moment of order p is finite only below α

$$E|X|^p < \infty \iff p < \alpha$$

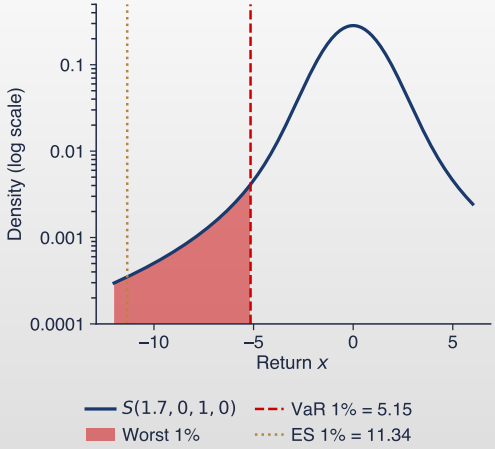
- $E|X|^p$: the mean of $|X|^p$; $p = 1$: mean, $p = 2$: variance
  - $1 < \alpha < 2$: finite mean, infinite variance; in S1 the mean is δ_1
  - $\alpha \leq 1$: neither mean nor variance
- The sample variance always exists

$$s_n^2 = \frac{1}{n-1} \sum_{t=1}^{n} (X_t - \bar{X})^2$$

- In the chart: s_n^2 as the sample grows
  - Normal: it settles at the true variance 2
  - stable, $\alpha = 1.7$: it keeps jumping upwards and does not converge

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Prerequisites for Today (10/14): VaR and ES



- Quantiles always exist; **VaR** (value at risk) at level 1%

$$\mathrm{VaR}_{1\%} = -q_{1\%}, \qquad P(X \le q_{1\%}) = 0.01$$

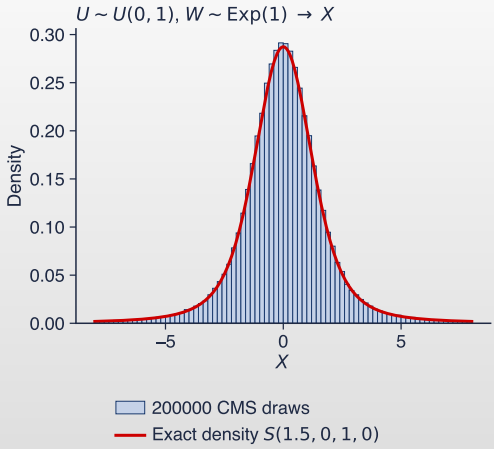
- $q_{1\%}$: the 1% quantile of the return; VaR: the loss exceeded on 1% of the days
- **ES** (expected shortfall): the mean loss beyond the VaR

$$\mathrm{ES}_{1\%} = -E\left[X \mid X \le q_{1\%}\right]$$

- a mean, so it needs $\alpha > 1$; for $\alpha \le 1$ it is infinite
  - in the chart, $S(1.7, 0, 1, 0)$: VaR 1% = 5.15, ES 1% = 11.34, more than twice the VaR
  - under the Normal distribution, ES 1% is only about 1.15 times VaR 1%

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Prerequisites for Today (11/14): Simulation by CMS



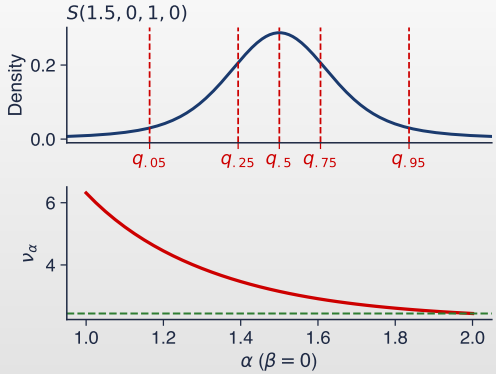
- **CMS** Chambers, Mallows & Stuck (1976) for $S(\alpha, 0, 1, 0)$: from a uniform and an exponential draw

$$X = \frac{\sin(\alpha V)}{(\cos V)^{1/\alpha}} \left(\frac{\cos((1-\alpha)V)}{W} \right)^{(1-\alpha)/\alpha}$$

- $U \sim U(0,1)$: uniform; $W \sim \mathrm{Exp}(1)$: exponential with mean 1, independent of U
  - $V = \pi(U - 1/2)$: a uniform angle on $(-\pi/2, \pi/2)$
  - then $\gamma X + \delta$ has scale γ and location δ
- Checks
  - $\alpha = 1$: the exponent is 0 and $X = \tan V$ (Cauchy)
  - in the chart: the histogram of 200 000 draws matches the exact density

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Prerequisites for Today (12/14): McCulloch's Quantile Method



- **McCulloch** McCulloch (1986): two ratios of sample quantiles

$$\nu_\alpha = \frac{q_{0.95} - q_{0.05}}{q_{0.75} - q_{0.25}}, \qquad \nu_\beta = \frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$$

- q_p: the sample quantile of order p (top panel)
  - ν_α: width of the tails relative to the body; it falls as α rises, to 2.439 for the Normal (bottom panel)
  - ν_β: asymmetry; $\nu_\beta = 0$ for a symmetric law
- Tables invert the ratios into $\hat{\alpha}$, $\hat{\beta}$; then the scale

$$\hat{\gamma} = \frac{q_{0.75} - q_{0.25}}{\phi_3(\hat{\alpha}, \hat{\beta})}$$

- $\phi_3(\alpha, \beta)$: the interquartile range of $S(\alpha, \beta, 1, 0)$; excerpt on the next slide

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Prerequisites for Today (13/14): McCulloch Tables (Excerpt)

ν_α	α
2.439	2.000
2.6	1.808
2.8	1.664
3.0	1.563
3.2	1.484
3.5	1.391
4.0	1.279
5.0	1.128
6.0	1.029

α	$\phi_{\mathbf{3}}(\alpha, 0)$
2.0	1.908
1.8	1.921
1.6	1.933
1.5	1.939
1.4	1.946
1.3	1.955
1.2	1.965
1.1	1.980
1.0	2.000

- Left: Table III, column $\nu_\beta = 0$; right: Table V, column $\beta = 0$ McCulloch (1986)
- Between two rows, interpolate linearly
- For $|\nu_\beta| < 0.05$ the $\beta = 0$ columns are close enough on paper

Prerequisites for Today (14/14): Maximum Likelihood, AIC and Bootstrap

- **ML** (maximum likelihood) Nolan (2001): the parameters that make the data most likely

$$(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\delta}) = \arg\max \ell, \qquad \ell = \sum_{t=1}^{n} \ln f(r_t; \alpha, \beta, \gamma, \delta)$$

- f: the stable density, computed numerically from φ; a hat marks an estimate
  - standard errors from the curvature of ℓ at its maximum; 95% CI: $\hat{\alpha} \pm 1.96\,\mathrm{SE}$
- **AIC** (Akaike information criterion) for comparing models

$$\mathrm{AIC} = 2k - 2\ell(\hat{\theta})$$

- k: the number of parameters (Normal 2, Student-t 3, stable 4); the lower AIC is preferred
- **Bootstrap**: resample the days with replacement B times and re-estimate each time
  - the standard deviation of the B estimates: the bootstrap SE; the 2.5% and 97.5% percentiles: a 95% interval

A1: The Scale of a Sum [Solved]

Task

- X_1, X_2 are independent copies of $S(\alpha, 0, 1, 0)$; $a = 3$, $b = 4$.
- 1. Compute c such that $3X_1 + 4X_2 \stackrel{d}{=} cX$ for $\alpha = 2$, $\alpha = 1.5$ and $\alpha = 1$.
- 2. For $\alpha = 2$, check the result with variances.
- 3. Explain why c grows as α falls.
- Report: three values of c and one sentence.

Solution

- 1. $c = (3^\alpha + 4^\alpha)^{1/\alpha}$: $\alpha = 2$: 5.00; $\alpha = 1.5$: 5.58; $\alpha = 1$: 7.00
- 2. $S(2, 0, 1, 0) = N(0, 2)$: $\mathrm{Var}(3X_1 + 4X_2) = (9 + 16) \cdot 2 = 50 = \mathrm{Var}(5X)$
- 3. With small α, large values are frequent and one term can dominate: the scale of the sum approaches the sum of the scales.

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A2: Horizons and Diversification [Proposed]

Solution discussed in class

Task

- Daily returns are i.i.d. $S(\alpha, 0, \gamma, 0)$. Model: A1.
- 1. Compute the scale of the 20-day sum, as a multiple of γ, for $\alpha = 2$, 1.7 and 1.5.
- 2. Compute the scale of an equally weighted portfolio of $N = 10$ i.i.d. assets for $\alpha = 2$, 1.5 and 1.
- 3. Explain what happens to diversification when $\alpha = 1$.
- Report: six multiples of γ and one sentence.

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A3: S1 and S0 at $t = 1$ [Solved]

Task

- X has $\alpha = 1.5$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$.
- 1. Compute $\varphi(1)$ in the S1 parameterisation.
- 2. Compute $\varphi(1)$ in the S0 parameterisation.
- 3. Explain the difference using the link between δ_0 and δ_1 (Prerequisites for Today (6/14)).
- Report: two complex numbers and one sentence.

Solution

- 1. $\tan \frac{3\pi}{4} = -1.0$;
  $\ln \varphi(1) = -[1 - i(0.3)(-1)] = -1 - 0.3i$;
  $\varphi(1) = e^{-1}(\cos 0.3 - i \sin 0.3) = 0.351 - 0.109i$
- 2. $|t|^{1-\alpha} - 1 = 0$ at $t = 1$: $\varphi(1) = e^{-1} = 0.368$
- 3. Same modulus e^{-1}; the laws differ by a shift: S1 with $\delta_1 = 0$ is S0 with $\delta_0 = 0.3 \cdot (-1.0) = -0.3$.

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A4: Special Cases and a Location Conversion [Proposed]

Solution discussed in class

Task

- Model: A3.
- 1. Write $\varphi(t)$ for $\alpha = 2$ and compare it with $\exp(i\mu t - \sigma^2 t^2/2)$, the characteristic function of $N(\mu, \sigma^2)$: what are μ and σ^2?
- 2. Write $\varphi(t)$ for $\alpha = 1$, $\beta = 0$ and name the law.
- 3. scipy reports $\alpha = 1.7$, $\beta = -0.2$, $\gamma = 0.6$, $\texttt{loc} = 0.05$; compute δ_0.
- Report: μ, σ^2, the name of the law and δ_0.

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A5: How Fast Do the Tails Fall? [Solved]

Task

- Daily losses have a power-law tail with $\alpha = 1.7$; a loss above x happens on 1 day in 50.
- 1. Compute $P(L > 2x)/P(L > x)$.
- 2. Compute how often a loss above $2x$ happens, and how often a loss above $4x$ happens.
- 3. Compare with a Normal law that has the same frequency of losses above x.
- Report: one ratio, two frequencies and one sentence.

Solution

- 1. $2^{-\mathbf{1.7}} = 0.308$
- 2. Above $2x$: 1 day in $50 \cdot 2^{\mathbf{1.7}} \approx 162$; above $4x$: 1 day in $50 \cdot 4^{\mathbf{1.7}} \approx 528$
- 3. Normal: $x = z_{\mathbf{0.98}}\sigma$; $P(L > 2x) = P(Z > 2z_{\mathbf{0.98}})$, i.e. 1 day in about 50 004: the power law keeps large losses frequent.

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A6: Tails and Moments for Other α [Proposed]

Solution discussed in class

Task

- A loss above x happens on 1 day in 100. Model: A5.
- 1. For $\alpha = 1.3$ and $\alpha = 1.9$, compute how often losses above $2x$ and above $3x$ happen.
- 2. For $\alpha = 1.9$, 1.3 and 0.8, say whether the mean and the variance exist.
- 3. Explain why a sample variance can be computed even when the variance does not exist.
- Report: four frequencies, a small table of moments and one sentence.

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A7: McCulloch from Five Quantiles [Solved]

Task

- Quantiles of 2000 daily returns (%): $q_{0.05} = -2.10$, $q_{0.25} = -0.55$, $q_{0.5} = 0.05$, $q_{0.75} = 0.62$, $q_{0.95} = 2.06$.
- 1. Compute ν_α and ν_β.
- 2. Read $\hat{\alpha}$ from the table excerpt (column $\nu_\beta = 0$).
- 3. Compute $\hat{\gamma}$ with Table V.
- Report: two ratios, $\hat{\alpha}$, $\hat{\gamma}$ and one sentence on the tails.

Solution

- 1. $\nu_\alpha = 4.16/1.17 = 3.556$; $\nu_\beta = (2.06 - 2.10 - 0.10)/4.16 = -0.034$, almost symmetric
- 2. Between $3.5 \mapsto 1.391$ and $4.0 \mapsto 1.279$: $\hat{\alpha} \approx 1.379$
- 3. $\phi_3(\hat{\alpha}, 0) \approx 1.948$, so $\hat{\gamma} = 1.17/1.948 = 0.601$
- With the full tables ($\nu_\beta \neq 0$): $\hat{\alpha} = 1.377$, $\hat{\beta} = -0.054$, $\hat{\gamma} = 0.599$; tails far heavier than for the Normal distribution ($\nu_\alpha = 2.439$).

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A8: McCulloch for a Crypto Asset [Proposed]

Solution discussed in class

Task

- Quantiles of daily returns (%): $q_{0.05} = -6.90$, $q_{0.25} = -1.45$, $q_{0.5} = 0.12$, $q_{0.75} = 1.75$, $q_{0.95} = 6.40$. Model: A7.
- 1. Compute ν_α and ν_β.
- 2. Read $\hat{\alpha}$ and compute $\hat{\gamma}$ from the table excerpts.
- 3. Compare with A7: which asset has the heavier tails?
- Report: two ratios, $\hat{\alpha}$, $\hat{\gamma}$ and one sentence.

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A9: One Stable Draw by CMS [Solved]

Task

- $\alpha = 1.5$, $\beta = 0$; a uniform draw $U = 0.8$ and an exponential draw $W = 1.2$.
- 1. Compute $V = \pi(U - 1/2)$.
- 2. Compute the three factors of the CMS formula.
- 3. Compute the draw X and the draw of $S(1.5, 0, 0.6, 0.05)$.
- Report: V, three factors, two draws.

Solution

- 1. $V = 0.3\pi = 0.942$
- 2. $\sin(1.5V) = 0.988$; $(\cos V)^{\mathbf{1/1.5}} = 0.702$; $(\cos(-0.5V)/1.2)^{\mathbf{-1/3}} = 1.104$
- 3. $X = 0.988/0.702 \times 1.104 = 1.554$; then $0.6X + 0.05$

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A10: CMS for the Normal and the Cauchy Laws [Proposed]

Solution discussed in class

Task

- Model: A9.
- 1. Show that for $\alpha = 2$ the CMS formula becomes $X = 2 \sin V \sqrt{W}$.
- 2. Show that for $\alpha = 1$ it becomes $X = \tan V$.
- 3. Compute both draws for $U = 0.8$, $W = 1.2$.
- Report: two short derivations and two numbers.

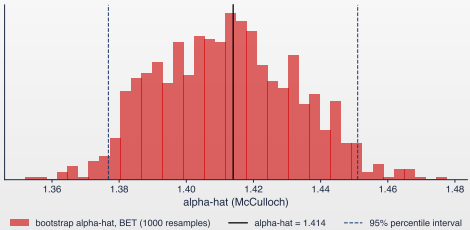
Back Exercise list

B1: How Heavy Are the Tails of the BET? [Solved]

- **Question**: how precisely does McCulloch's quantile method estimate the stable law of daily BET returns?
- **Inputs**: BET closes, 2000–2026, daily log returns in %
- **Tasks**
  1. Compute the 5%, 25%, 50%, 75% and 95% quantiles and the ratios ν_α and ν_β.
  2. Compute the McCulloch estimates $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$ and $\hat{\delta}_0$ with the full tables.
  3. Bootstrap $\hat{\alpha}$ with 1000 resamples of the days and report the standard error and the 95% percentile interval.
  4. Interpretation: is $\alpha = 2$ (the Normal distribution) compatible with the data?
- **Report**: five quantiles, two ratios, four estimates, the interval and one sentence
- Notebook: section B1

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B1: Solution [Solved]



- $n = 6\,670$; quantiles -1.93, -0.50, 0.08, 0.66, 2.06; $\nu_\alpha = 3.425$, $\nu_\beta = -0.006$
- $\hat{\alpha} = 1.414$, $\hat{\beta} = -0.011$, $\hat{\gamma} = 0.60\%$, $\hat{\delta}_{\mathbf{0}} = 0.079\%$; bootstrap SE 0.021, 95% interval $[1.377, 1.451]$
- Interpretation: $\alpha = 2$ lies far outside the interval; the BET body and tails are much wider than the Normal distribution allows

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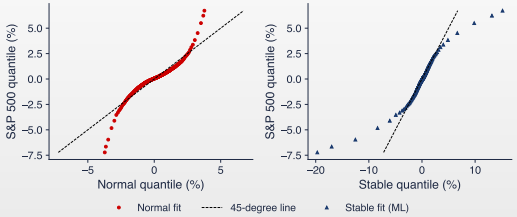
B2: Three More Markets [Proposed]

- **Question**: which of the S&P 500, DAX and Bitcoin has the heaviest tails by McCulloch's method? Model: B1.
- **Inputs**: S&P 500 and DAX since 2000, Bitcoin since 2014, daily log returns in %
- **Tasks**
  1. Repeat steps 1–3 of B1 for each series.
  2. Put ν_α, $\hat{\alpha}$ with its interval, $\hat{\beta}$ and $\hat{\gamma}$ of the four series (with the BET) in one table.
  3. Interpretation: is the difference between the $\hat{\alpha}$ of Bitcoin and of the S&P 500 larger than the estimation error?
- **Report**: one table and two sentences
- Notebook: section B2

B3: Maximum Likelihood for the S&P 500 [Solved]

- **Question**: does the ML stable fit describe the S&P 500 better than the Normal distribution, in the body and in the tails?
- **Inputs**: S&P 500 closes since 2000, daily log returns in %
- **Tasks**
  1. Fit the stable law by ML (S0) and report $\hat{\alpha}$ and $\hat{\beta}$ with standard errors, $\hat{\gamma}$, $\hat{\delta}_0$ and $\hat{\delta}_1$.
  2. Compare $\hat{\alpha}$ with the McCulloch estimate.
  3. Fit the Normal and the Student-t distributions and compare the AIC (Prerequisites for Today (14/14)) of the three models.
  4. Draw the QQ plot of the data against the Normal and the stable fits.
  5. Interpretation: where does each model fail?
- **Report**: the estimates, three AIC values, the QQ plot and two sentences
- Notebook: section B3

B3: Solution [Solved]



- ML: $\hat{\alpha} = 1.540$ (SE 0.021, 95% CI $[1.499, 1.581]$), $\hat{\beta} = -0.181$ (0.036), $\hat{\gamma} = 0.58$, $\hat{\delta}_{\mathbf{0}} = 0.093$, $\hat{\delta}_{\mathbf{1}} = 0.000$; McCulloch $\hat{\alpha} = 1.438$
- AIC: Normal 21 650, Student-t 19 702, stable 19 794: the stable law beats the Normal fit by 1 856 points, the Student-t beats the stable law by 92
- Interpretation: the Normal fit misses the tails (0.1% quantile -3.72 vs -7.22 in the data); the stable fit overshoots them (-19.70)

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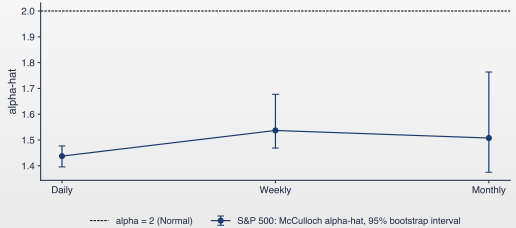
B4: Large Bitcoin Losses under Three Models [Proposed]

- **Question**: which model predicts the number of large daily Bitcoin losses best? Model: B3.
- **Inputs**: Bitcoin closes since 2014, 7 days a week, daily log returns in %
- **Tasks**
  1. Fit the Normal, Student-t and stable (ML) laws.
  2. Count the days with a loss above 10% and above 20%.
  3. Compute the expected number of such days under each model, $n \times P(r < -x)$.
  4. Compute VaR 1% of the data and of each model.
  5. Interpretation: which model would you use to set the margin of a Bitcoin position? Justify the choice.
- **Report**: one table of counts, four VaR values and two sentences
- Notebook: section B4

B5: Does α Stay the Same When Returns Are Summed? [Solved]

- **Question**: if S&P 500 daily returns were i.i.d. stable, weekly and monthly returns would have the same α; do they?
- **Inputs**: S&P 500 daily log returns since 2000; weekly and monthly sums
- **Tasks**
  1. Compute the weekly and monthly log returns as sums of the daily ones.
  2. Compute the McCulloch $\hat{\alpha}$ at the three horizons.
  3. Bootstrap each $\hat{\alpha}$ with 500 resamples and report the 95% intervals.
  4. Interpretation: is the pattern consistent with i.i.d. stable returns?
- **Report**: three estimates with intervals, the chart and two sentences
- Notebook: section B5

B5: Solution [Solved]



- Daily ($n = 6\,714$): 1.44 [1.40, 1.48]; weekly (1 394): 1.54 [1.47, 1.68]; monthly (321): 1.51 [1.37, 1.76]
- $\hat{\alpha}$ is higher for weekly than for daily returns; the monthly interval is wide and reaches up to 1.76
- Interpretation: a rising α is a warning sign against i.i.d. stable returns; monthly samples are too short for a firm answer (Lecture 3 repeats the test with ML)

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B6: Does the Sample Variance Settle? [Proposed]

- **Question**: is the sample variance of BET and Bitcoin returns as large and as unstable as a stable law with the fitted α implies? Models: B1, B5.
- **Inputs**: BET since 2000, Bitcoin since 2014, daily log returns in %
- **Tasks**
  1. Compute the sample variance of each series over the full sample and over its first half.
  2. Simulate 200 paths of the same length from the McCulloch fit (CMS) and compute the sample variance of each path.
  3. Report the median and the 5% and 95% percentiles of the simulated variances, and the share below the observed one.
  4. Interpretation: what does the comparison say about the infinite-variance hypothesis?
- **Report**: a table with two rows and one sentence
- Notebook: section B6

C1: Is α Constant over Time? [Proposed]

- **Question**: do the tails of the BET and the S&P 500 change from one five-year period to the next?
- **Inputs**: daily log returns since 2000, split into 2000–2004, 2005–2009, 2010–2014, 2015–2019, 2020–2026; models: B1, B3
- **Tasks**
  1. Compute the McCulloch $\hat{\alpha}$ with a bootstrap interval in each period, for each index.
  2. Say which periods differ significantly.
  3. Propose one explanation for the changes and one way to test it (for example, returns divided by a rolling volatility).
  4. Interpretation: what would a changing α mean for a risk model estimated on old data?
- **Report**: a table, a short list of significant changes and a plan for a project
- Notebook: section C1

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C2: Audit an AI Answer [Proposed]

- **A student asked an AI assistant to fit a stable law to the daily S&P 500 log returns (%) since 2000. The answer:**
- `(a) scipy's levy_stable uses S0 by default, so loc = 0.000 is the S0 location delta0.`
- `(b) With alpha = 1.540 and gamma = 0.58, the variance of daily returns is 2 gamma^2 = 0.68.`
- `(c) The scale of 20-day returns is sqrt(20) x gamma = 2.60%.`
- `(d) Since alpha < 2, the VaR 1% of this model does not exist.`
- `(e) Since alpha > 1, the mean daily return exists.`
- **Tasks**
  1. **For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).**
- **Report: a list of five verdicts with one line of justification each.**

What You Should Take from Today

- A stable sum of n terms scales like $n^{1/\alpha}$, faster than $\sqrt{n}$ when $\alpha < 2$
- Know the parameterisation: S0 and S1 differ in the location; scipy uses S1
- McCulloch: five quantiles and a table; ML: more precise; bootstrap for intervals
- Real returns: $\hat{\alpha}$ well below 2 at the daily horizon, yet the variance settles and $\hat{\alpha}$ rises with aggregation
- An AI answer is a draft: check the parameterisation, the scale rule and what exists for $\alpha < 2$

After the Seminar

- Lecture 3 develops each topic of today: stability, the generalised CLT, parameters, tails, simulation, estimation and the critique
- C1 can grow into a team project: rolling windows, ML, more markets
- Reading: Nolan (2020), Ch. 1; Borak, Härdle & Weron (2005); McCulloch (1986)

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